Journal of Sports Sciences

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/rjsp20

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To cite this article: Julie A. Hides, Daniel A. Brown, Luke N. Anning, Peter D. Blanch & Felix T. Leung (04 Dec 2024): Decreased isometric neck strength is a risk factor for head, neck and face injuries in professional rugby league players, Journal of Sports Sciences, DOI: 10.1080/02640414.2024.2432793

To link to this article: https://doi.org/10.1080/02640414.2024.2432793

Published online: 04 Dec 2024.

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JOURNAL OF SPORTS SCIENCES
https://doi.org/10.1080/02640414.2024.2432793

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Decreased isometric neck strength is a risk factor for head, neck and face injuries in professional rugby league players

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ABSTRACT

Head, neck and face injuries are a concern in contact sports. This exploratory study aimed to establish 1) injury risk factors for head, neck and face injuries and 2) clinical cut-off values related to strength, endurance and proprioception of the cervical spine in a team of professional rugby league players. Pre-season assessments of isometric strength of the flexor, extensor and lateral flexor muscles, endurance of the flexor muscles and joint position error were conducted. Injuries resulting in games missed were recorded. Cross-tabulations were used to determine the unadjusted odds ratios for the measures as risk factors for playing season injuries. The unadjusted odds ratio (OR) values indicated that if a player had weaker extensors of the neck (<36.4 kg, $p = 0.014$; <3.4N/kg, $p = 0.014$) or asymmetry of isometric strength of their lateral flexor muscles (left-to-right ratio <0.91, $p = 0.005$), their odds of games missed due to season head, neck and face injuries was increased (OR extensors = 8; 95% CI = 1.5-42.5 OR asymmetry of lateral flexor muscles OR = 12.6; 95% CI = 2.0-79.4). As muscle strength is modifiable, the clinical application of this study would involve targeting players in the team beneath the clinical cut-off value.

ARTICLE HISTORY
Received 12 July 2024
Accepted 14 November 2024

KEYWORDS
Cervical spine; endurance; football; injury prevention; proprioception

Introduction

Head, neck and face injuries are common in contact sports. These include concussion, which is the most common diagnosis of emergency department visits for American Football-related head and neck injuries (Mehta et al., 2022) and the most common injury in professional rugby union (Chavarro-Nieto et al., 2021). These injuries are a major public health concern, with concussion linked to long-term sequalae effects on athlete health (Patricios et al., 2023). In rugby league, the risk of players sustaining injuries to the head and neck region is high due to the intensity and physical nature of the game (Gardner et al., 2014). A recent systematic review reported that while the overall incidence of rugby league match and training injuries had decreased since 2014, there had been a tenfold increase in the incidence of injuries to the head and neck region (RR: 10.7 [95% CI: 10.1–11.3]; $p < 0.001$) (King et al., 2022). The study reported that most of these injuries were sustained by the ball carrier and tackler and concluded that further injury prevention interventions are needed (King et al., 2022).

In 2017 and 2018, in the Australian National Rugby League (NRL) competition, it was reported that one concussion occurred approximately every 2.70 games (Iverson & Gardner, 2022). In the 2021 NRL season, concussion injuries accounted for 37% of injuries, resulting in games missed (O'Connor, 2022). Facial injuries also pose a large health burden for athletes, potentially resulting in time loss and surgery, and there is a paucity of the literature examining sport-specific risk factors (Black et al., 2017). Injuries to the neck region are also common in rugby league, and there is the potential for accumulated microtrauma to the neck through "whiplash" type injuries (Hoskins et al., 2006). Fortunately, acute cervical spinal cord injuries are rare events, but when they do occur, they are usually the result of being tackled (Hoskins et al., 2006). As head, neck and face injuries can have serious consequences for the athlete, it is important to identify modifiable risk factors for these injuries.

Considering that the most common mechanism of injury for sustaining head, neck and face injuries in rugby league and rugby union is tackling (or being tackled), it has been hypothesized that the strength and endurance of neck muscles should be considered. It has been demonstrated that male rugby players with a history of concussion had altered strength of their neck muscles (Nutt et al., 2022). It has further been proposed that stronger neck muscles may be able to mitigate the whiplash-like movements that can occur during impact (Nutt et al., 2022). Furthermore, during tackling, optimal spinal positioning in preparation for impact is believed to be important to allow force dissipation from the point of contact at the neck and shoulder and through the trunk (Seminati et al., 2017). Recent studies conducted in adult rugby union players have identified associations between decreased neck strength and impaired proprioception of the neck and increased risk of sustaining a concussion (Farley et al., 2022a & 2022b). Increased joint

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position error (indicating poorer proprioception) has been shown to significantly increase the odds of sustaining a head or neck injury in rugby league, rugby union, and Australian Football League players (Hides et al., 2017). If players cannot position themselves well and preset the head in an optimal position before contact (Pinsault et al., 2010), this could help to explain the increased risk of head/neck injuries associated with altered cervical proprioception.

The researchers who conducted the current investigation were approached by a professional rugby league team to determine if (weaker) neck muscle strength, (decreased) endurance and (poorer) proprioception were related to head, neck and face injuries in their club. If relationships were found to be present, the medical team were interested in determining clinical cut-off values appropriate for the rugby league players at their club so that they could provide interventions for the players with increased risk. The aims of this study were therefore to examine for injury risk factors and establish clinical cut-off values related to clinical measures of strength, endurance and proprioception of the cervical spine in a team of professional rugby league players.

Materials and methods

Participants

Players aged 18 years and over, who were part of a professional rugby league squad in Queensland, Australia, in 2023, were eligible for participation in this prospective observational study. Players were excluded from the pre-season testing for the study if they had sustained a recent concussion (within 1 month), reported any injury to the cervical spine where they were not medically cleared to train or play, or if they had any pain in the cervical spine region greater than 2/10 on a visual analogue scale at rest or during the warmup prior to neck strength testing. Thirty-two male rugby league players participated in the study. Mean (SD) participant age (yrs), height (cm), and mass (kg) were 23.9 (3.5) yrs, 174 (7.1 cm), and 83 (9.0) kg. Written informed consent was obtained for all participants. The study was approved by the institutional Human Research Ethics Committee and was performed in line with the principles of the Declaration of Helsinki.

Assessment of cervical spine function

To assess the proprioception of the cervical spine, a joint position error test was used (Farley et al, 2022a). Participants were seated on a chair placed 90 cm away from a standardised target, with a laser pointer placed on their forehead (Figure 1a). With the laser beam pointing to the centre of the target, participants were instructed (first with their eyes open) to move their head as far as they comfortably could in each direction and then to return the laser beam to the centre of the target. After being familiarised with the task, participants were blind folded for the test, which was repeated three times for the directions of cervical spine flexion, extension and rotation (to both sides). Participants indicated when they thought they had returned to the starting position, and this point was marked by a second assessor on the target with a pen. The tester manually repositioned the participant's head with the laser beam on the centre of the target (starting position) before each trial. The Absolute Error (AE) was recorded.

To test the endurance of the cervical flexor muscles (Olson et al., 2006), participants were first provided with an explanation and a demonstration of a "chin tuck" manoeuvre in a sitting position. For testing, they were placed in a supine crook lying position (with their arms by their sides). Participants performed a craniocervical flexion (chin tuck) movement followed by lower cervical flexion by raising their head to allow the tester to slide their index and middle fingers under the participant's head (at the most posterior aspect of the occiput) (Figure 1b). The duration for which they could hold this position was timed. The test was concluded when one of the four criteria was met. These included when the participant: 1) experienced pain and was unwilling to continue, 2) held the test position, but reached the end of their endurance and was unwilling to continue, 3) could not maintain the chin tuck position, or 4) raised their head (while maintaining a chin

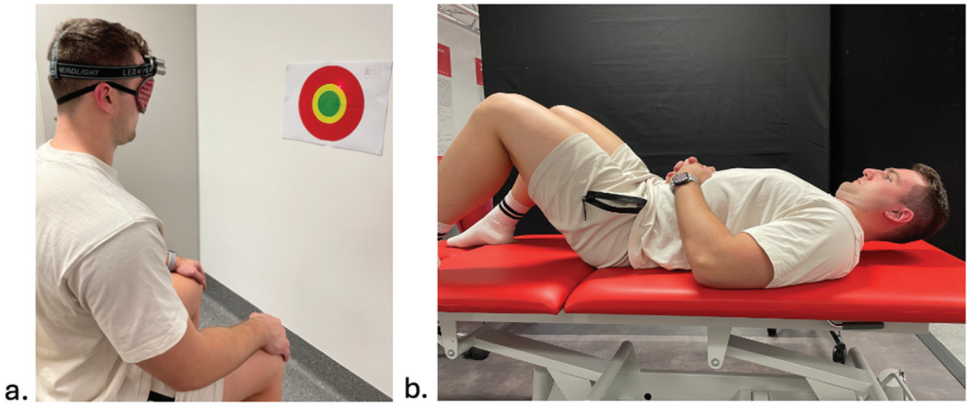

Figure 1. Assessment of cervical spine function including a) proprioception using the joint position error test, and b) flexor muscle endurance.

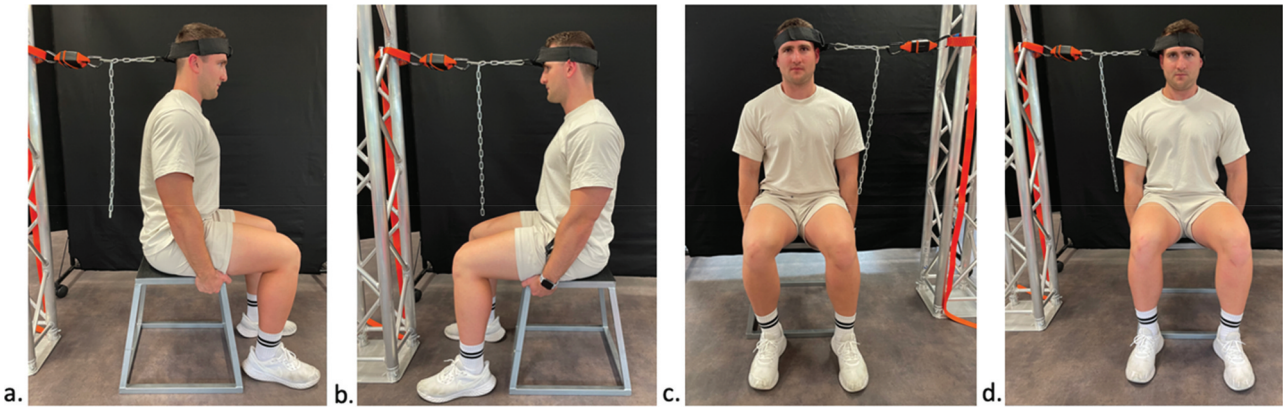

Figure 2. Assessment of maximal isometric neck muscle strength using a pull dynamometer to examine; a) flexion, b) extension, c) right lateral flexion, and d) left lateral flexion strength.

tuck position) such that they no longer maintained contact with the tester's fingers.

Assessment of neck muscle strength was conducted using a validated and reliable protocol (Fuller et al., 2022). Maximal isometric neck strength was assessed for the directions of flexion, extension and lateral flexion (left and right) of the cervical spine using a pull dynamometer (K-Pull KINVENT, Montpellier, France) (Figure 2). Participants were seated in an upright position on a 45-cm-high bench, with their feet positioned hip width apart and flat on the floor, and the arms placed by their sides holding onto the seat. The head was positioned in a neutral position during the testing for all directions. A warm-up was conducted prior to testing, with participants completing self-resisted isometric contractions in all four directions at 50–70% of their maximal contraction. A harness was placed around the head above the ear line and across the frontal and parietal bones. For the testing, participants were instructed to produce "a maximal effort as quickly as possible" while ensuring their neck was "as stiff as possible to maintain a neutral position". For each direction, three repetitions of 3-second contractions were recorded with 30-second rest between each repetition. Participants were given a countdown before the verbal cue of "pull, pull, pull, pull, pull and relax" was given. The maximal contraction in kilograms (kg) was recorded. Strength was also normalized to body weight (Newtons/Kilogram, N/kg). Prior to commencement of the study, an intra-rater reliability study was conducted on 20 young adults (male and female) independent of the current investigation. Results showed that intra-class correlation co-efficients (ICCs) for isometric strength of the cervical flexor, extensor and lateral flexion muscles were good to excellent for test–retest reliability, ranging from 0.88 to 0.94 (standard error of measurements, SEMs, range 1.5–2 kg). These results were similar to the previous results of Fuller et al. (2022), where ICCs for these strength measures ranged from 0.86 to 0.92.

Season injury data

Demographic information and history of previous concussions sustained by players in the last 12 months prior to commencement of the study were provided by the club medical team. Injury data were collected by the medical team during the pre-season and 2023 playing season, following the collection of the neck strength, endurance and joint position error data in the pre-season, and were provided to the research team at the end of the playing season. An injury was defined as a physical condition related to playing or training for football that prevented a player from completing a game or playing in subsequent games. Head, neck and face injuries were diagnosed and recorded by medical staff

Statistical analysis

SPSS version 28.0 (IBM) was used for analyses. Data were examined for outliers and normal distribution using summary statistics, histograms, normality plots and the Shapiro–Wilk test. Prior to conducting the main analyses, a preliminary analysis was conducted to examine for any relationship between a history of concussion in the last 12 months and the variables of interest, using one-way ANOVAs for variables that satisfied the assumption of normality. A Mann-Whitney U-test was used for those variables that did not. This preliminary analysis was performed to check for confounders that could possibly affect the relationship between the measures of the cervical spine and injury risk. The measurements tested as risk factors included continuous variables (age, height, mass and measures of strength and endurance of the cervical spine muscles and joint position error). In the main analysis, all factors were assessed against head, neck and face injuries in the 2023 pre- and playing season. Continuous variables of height and mass were analysed using one-way ANOVAs, with the between-group factor being the injury variable ("head neck and face" or "no head, neck or face injuries"). One-factor repeated-measures ANOVAs were conducted to compare the continuous variables which were normally distributed. For those variables which did not satisfy the assumption of normality, a Mann-Whitney U-test compared the differences between the two independent injury groups. Separate analyses were conducted for each dependent variable. Because this was an exploratory study aimed at identifying the priority for subsequent research, effect sizes (mean difference/pooled standard deviation) >0.5 and liberal p values <0.1 were considered to help identify

clinically meaningful effects (Bender & Lange, 2001). If a significant relationship existed, receiver operating characteristic (ROC) curves were used to determine an optimal cut point for each continuous variable in predicting the injury outcome variables. The optimal cut point was obtained as the point where the true-positive rate (sensitivity) was maximized and the false-positive rate (1 – specificity) was minimized, that is, the point closest to the top of the y-axis. The cut points were then used to convert the continuous variables to a binary form. The sensitivity and specificity of each measure and the unadjusted odds ratios (OR) for predicting injuries and head, neck and face injuries in the pre- and playing season were estimated from cross-tabulations. Because this was an exploratory study, no adjustments for multiple comparisons were made, also to avoid Type II errors (Bender & Lange, 2001). As the study was conducted in a clinical setting with a pre-determined sample size (players from one professional team), retrospective analyses were conducted using G*Power 3.1.9.4 to establish the power of the study.

Results

History of concussion in the last 12 months

Of the 32 players included in the current investigation, 5 had sustained a head, neck or face injury in 2022. Results of the preliminary between-group analysis comparing those with and without a history of concussion showed that only one of the factors examined in the current investigation was significantly different. The five players with a history of head, neck and face injuries had stronger lateral flexors (to the left side) than those without a history of injury ($p < 0.05$), but there was not a significant between-group difference observed for the left-to-right lateral flexion strength ratio.

Number of pre- and playing season injuries

In the 2023 pre- and playing season, there were 11 head, neck and face injuries (34% of the team). Of these, five were concussion injuries, four were neck injuries and three were injuries to the face. There were no significant differences between the "head, neck or face injury group" or the "no head, neck or face group" for age, height or mass (all $p > 0.05$).

Pre-season and season injuries and neck muscle endurance, strength and joint position error

Pre-season and season injuries were related to the strength of the cervical spine muscles (Table 1) but not associated with measures of endurance of the deep cervical flexor muscles or joint position error in the sample of players tested. Players who sustained a head, neck or face injury had weaker cervical spine extensor muscles ($p = 0.02$, ES = 0.9, for N/kg; $p = 0.008$, ES = 1, for Kg) and lateral flexor muscles (lateral flexion right N/kg $p = 0.046$, ES = 1). Players who suffered a head, neck or face injury in the pre-season or playing season also had more asymmetry of strength of their lateral flexor muscles ($p < 0.001$, ES = 1.5). Measures of strength of the cervical spine flexor muscles and

Table 1. Strength, endurance and joint position error measures of the cervical spine for rugby league players who did or did not sustain head, neck or face injuries in the 2023 pre- and playing season.

		No head, neck or concussion injuries during 2023 playing season	Head, neck or concussion injury during 2023 playing season (n = 11)		
Variable	N	Mean, SD; (95% CI)	Mean, SD; (95% CI)	P value	Effect Size
Endurance (sec)	32	64.9, 31.3; (50.7-79.1)	62,21.5; (47.5-76.4)	0.737	0.1
Strength flexors (Kg)	31	30.6, 6.6; (27.5-33.7)	31.2, 9.2; (25-37.5)	0.84	0.1
Strength flexors (N/Kg)	31	3.1, 0.7; (2.8-3.4)	3.1,0.8; (2.5-3.6)	0.994	0
Strength LF Left (Kg)	31	28.5, 6.0; (25.7-31.3)	28.2, 4.5 (25.2-31.2)	0.878	0.1
Strength LF Right (Kg)	31	27.7, 6.0; (24.9-30.6)	23.8, 5.0; (20.4-27.1)	0.073	**0.7**
Strength LF Left (N/Kg)	31	2.9, 0.6; (2.6-3.2)	2.8, 0.3; (2.6-3)	0.695	0.2
Strength LF Right (N/Kg)	31	2.8, 0.6; (2.5-3.1)	2.4, 0.4; (2.1-2.7)	**0.046**	**1.0**
Strength LF ratio	31	0.98, 0.1; (0.93-1)	0.84, 0.1; (0.78-0.9)	**<0.001**	**1.5**
		Median, IQ range; (95% CI)	Median, IQ range; (95% CI)	P value	Hedge's g
Strength extensors (Kg)	31	41.1,12.6; (36.6-43)	31.3, 9.1; (29.9-37.7)	**0.02**	**0.9**
Strength extensors (N/Kg)	31	4.1, 1.1; (3.7-4.3)	3.2, 0.6; (3-3.7)	**0.008**	**1**
Strength Flex/Ext ratio	31	0.8, 0.2; (0.74-0.84)	0.86, 0.5; (0.76-1.1)	0.183	**0.8**
Left Rotation error (deg)	32	2.8, 1.1; (2.4-3.7)	2.1, 2.7; (1.8-3.6)	0.506	0.2
Right Rotation error (deg)	32	2.4, 1.6 (2.1-3.2)	2.2, 1.6; (1.9-3.4)	0.969	0.02
Extension error (deg)	32	3.1, 1.5; (2.5-3.8)	2.9, 1.7; (2-3.6)	0.938	0.2
Flexion error (deg)	32	2.8, 1.3; (2.2-3.8)	2.8, 1.3; (2.3-3.5)	0.481	0.1
JPE absolute average(deg)	32	2.8, 0.8; (2.4-3.5)	2.8,1; (2.3-3.3)	0.969	0.2

Sec = seconds, *kg* = kilograms, *LF* = lateral flexion, *N/kg* = Newtons/kilogram, *Flex/Ext* = flexion/extension, *SD* = standard deviation, *ES* = effect size, *IQ range* = interquartile range, 95% CI = 95% confidence interval.

Table 2. Results of ROC curves with optimal clinical cut-off points for variables associated with sustaining a head, neck or face injury in the 2023 pre-season and playing season.

Measure	AUC	95% CI	P value	Optimal risk cut point	Sensitivity	Specificity
Strength LF R (N/kg)	0.75	0.6, 0.9	0.023	2.5 kg	70%	73%
Strength LF ratio	0.86	0.7, 1.0	0.001	0.91	80%	82%
Strength extensors (kg)	0.76	0.6, 0.9	0.021	36.4 kg	75%	73%
Strength extensors (N/kg)	0.78	0.6, 1.00	0.01	3.4 N/kg	80%	64%

LF = lateral flexion, *R* = right, kg = kilogram, N/kg = Newtons/kilogram, CI = confidence interval.

Table 3. Unadjusted associations between neck muscle strength risk factors and head, neck and concussion injuries in the pre- and playing season in rugby league players.

Measure	% "non-risk" group with injuries in the season	%"risk" group with injuries in the playing season	Unadjusted odds ratio	P value
Strength LF R (N/kg)	20.0%	50.0%	4 (0.8,19.8)	0.085
Strength LF ratio	12.5%	64.3%	12.6 (2,79.4)	0.005
Strength extensors (kg)	16.7%	61.5%	8 (1.5,42.5)	0.014
Strength extensors (N/kg)	16.7%	61.5%	8 (1.5,42.5)	0.014

LF = lateral flexion, *R* = right, kg = kilogram, N/kg = Newtons/kilogram, CI = confidence interval.

the ratio of cervical flexor to extensor muscle strength were not significantly different between groups. Measures with a significant area under the curve (AUC) are shown in Table 2. The measure with the highest sensitivity and specificity for pre- and in-season injuries was the lateral flexion strength ratio. The unadjusted odd ratios (ORs) for the risk factors associated with pre- and playing season injuries are shown in Table 3. The unadjusted OR values in Table 3 indicate that if a player had weaker lateral flexor (<2.5 N/kg) or extensor muscles in the cervical spine (<36.4 kg or <3.4N/kg) or increased asymmetry of strength of the lateral flexor muscles, their odds of a head, neck or face injury during the pre- and playing season increased significantly. For decreased strength and asymmetry of lateral flexion of the cervical spine, the clinical cut-off values identified the same players as being "at risk". Likewise, the raw strength of the extensor muscles and the normalised to body-weight variable (N/kg), identified the same "at risk" players.

Post-hoc power analysis for a difference between means for two independent groups was conducted using effect sizes for each dependent variable and an alpha value of 0.05. The power (1 − β error probability) for the variables of strength of lateral flexion right (kg) was 0.44; strength of lateral flexion right (N/kg) was 0.74; lateral flexion strength ratio was 0.97; strength of the extensor muscles (kg) was 0.97 and strength of the extensors (N/kg) was 0.74.

Discussion

The main findings of the current investigation were that players in the team studied with weaker cervical spine extensor muscles (<36.4 kg; $p = 0.014$; <3.4 N/kg; $p = 0.014$) or greater asymmetry of strength of the lateral flexor muscles (left-to-right ratio < 0.91, $p = 0.005$) at the start of the pre-season had increased odds of sustaining season head, neck and face injuries (OR extensors = 8; 95% CI = 1.5–42.5; OR asymmetry of lateral flexor muscles OR = 12.6; 95% CI = 2.0–79.4) in the pre-season and playing season. Clinical cut-offs for these variables for the rugby league players studied were determined.

While it has been previously reported that athletes with greater neck strength have a lower magnitude of head impacts across a variety of sports (Collins et al., 2014; Elliott et al., 2021; Nutt et al., 2022), there is conflicting evidence regarding whether increasing the strength or endurance of neck muscles can mitigate the risk of head, neck and face injuries. For example, while a study of high-school basketball, lacrosse and soccer athletes (Collins et al., 2014) showed that greater strength of neck muscles *was* associated with a lower incidence of sport-related concussion, another study reported that there was not an association between neck endurance of the deep cervical spine flexor muscles and the incidence of sport-related concussion injuries in university basketball, hockey and soccer players (Baker et al., 2019). A recent systematic review and meta-analysis examining the relationship between neck muscle strength and sports-related concussion concluded that there was very low-certainty evidence of a small but non-significant relationship between neck strength and the incidence of sports-related concussion injuries. This was proposed to be due to a paucity of research available and the high heterogeneity between studies (Garrett et al., 2023). Another systematic review and meta-analysis reported that intervention programmes that included neck strengthening were effective at decreasing the incidence of sports-related concussion (relative risk 0.54 (95% CI 0.50–0.95)), suggesting that neck strengthening intervention programmes can reduce the incidence of SRC in an athletic population (Versteegh et al., 2021). A longitudinal study of 225 professional male rugby union players reported that a 10% increase in strength of the neck extensor muscles across the playing season was associated with a 13% decrease in concussion injuries (Farley et al., 2022b). While the method of assessing neck muscle strength was different to that used in the current investigation (handheld dynamometer rather than a harness), it has been reported that the concurrent validity of handheld dynamometry for measuring isometric strength of neck muscles in rugby players was acceptable when compared with mounted dynamometers, and both methods were shown to be reliable (Ashall et al., 2021). The values reported for the strength of extensor muscles (Farley et al., 2022b) were similar to those reported in our study for players who did not suffer a head, neck or face injury (42.5 kg, SD 1.3) and somewhat higher than results of the current investigation for those who did suffer a head, neck or face injury (38.2 kg, SD 1.2). The values for the strength of the flexor and lateral flexor muscles

in the study by Farley et al. (2022b) were similar to the results obtained in the current investigation, with the cervical extensor muscles being the strongest, followed by cervical flexor muscles, followed by lateral flexor muscles. This similarity in profile of muscle strength across directions of cervical spine flexion, extension and lateral flexion, and the demonstrated association between greater strength of the cervical extensor muscles and concussion in this longitudinal study (Farley et al., 2022) may indicate that strengthening of the cervical extensor muscles could warrant further investigation.

The lack of an association between endurance of the cervical spine flexor muscles and head, neck and face injuries in the current investigation supports the previous results of Baker et al. (2019). It has been proposed that a high rate of force development may be required by muscles of the cervical spine to decrease the risk of sport-related concussions, especially when the mechanism involves a rapid whiplash-type movement (Elliott et al., 2021). Because the most common mechanism of head, neck and face injuries in rugby league is tackling (or being tackled), it is possible that endurance of neck muscles may be less related to sports related head and neck injuries, where a rapid reaction and contraction time of the cervical spine muscles may be required (Elliott et al., 2021).

The results of the current investigation for joint position error showed that the players in the team tested demonstrated less error than has been reported in previous studies for both those who did and did not go on to sustain a head, neck or face injury (Hides et al., 2017). Greater joint position error of the cervical spine has previously been found to be associated with an increased risk of concussion injuries in rugby league, union and Australian Football players when the error in rotation was greater than 4.5 (SD 1.7) degrees, which is a greater joint position error than was observed in the current investigation.

Limitations

The main limitation was that a sample of convenience (players from a single NRL club) was assessed in the current investigation. A post hoc power analysis showed that the study was adequately powered (>0.8) for the dependent variable of ratio of the lateral flexor muscles and slightly underpowered for strength of the cervical extensor muscles in N/kg and strength of lateral flexion right in N/kg (power for both = 0.74). The study was not adequately powered (<0.8) for the dependent variable of strength of the cervical spine extensor muscles in Kg (power = 0.65). Based on these results, using an a priori estimation of sample size, for the strength of the cervical spine muscles in extension in N/kg and lateral flexion right in N/kg, 33 players without a head, face or neck injury and 11 with the injury would be required. For the variable of cervical extension in Kg, 40 players without a head, face or neck injury and 14 with the injury would be required. Due to the small sample size in the current investigation, results should therefore be interpreted with caution. Furthermore, the wide confidence intervals of the odds ratios in the current exploratory investigation are likely to be related to the small sample size and may indicate that while the factors identified can be considered as increasing the likelihood of injury, their precision may be limited. The cut point values determined in this study may not be transferable to other rugby league teams, to female players or other sports. Future studies with larger sample sizes and players from several NRL clubs would allow a more thorough investigation of a greater number of potential risk factors.

Conclusion

While there is currently conflicting evidence regarding the possible protective effects of increasing the strength of cervical spine extensor muscles in contact sport athletes, the results of the current investigation are promising as players from the team tested with strength deficits had increased odds of suffering head, neck and facial injuries in the pre-season and playing season following testing. The finding of an association between deficits in between side symmetry of lateral flexor muscles of the cervical spine and increased odds of head, neck and facial injuries is a novel finding which has not been previously reported. While studies with larger sample sizes are required to confirm the findings of the current investigation, results suggest that intervention studies targeting strengthening of these muscles in players identified as having decreased strength may be warranted.

Practical implications

- This study represents a first step in exploring the relationship between the strength of neck muscles and head, neck and face injuries in professional rugby league players. As players in the team with weaker extensor muscles of the cervical spine had higher odds of head, neck and face injuries, exercise strategies targeting strengthening of the extensor muscles could be considered in these players.
- Increasing the symmetry of strength of lateral flexor muscles may decrease the odds of sustaining head, neck and face injuries in rugby league players in those with asymmetry.
- Establishing clinical cut-off values for the strength of cervical extensor and lateral flexor muscles for rugby league players in the team represents a first step towards identifying players who may possibly benefit the most from undertaking a targeted strength training.
- Further studies are required to determine the ideal method of strength training for those involved in contact sports. Of clinical importance, the ideal method of strengthening neck muscles at this time is unknown. A nuanced approach incorporating specific and targeted multi-planar training may be required.

Acknowledgments

We would like to acknowledge the players from the Brisbane Broncos Rugby League Football Club Ltd, Brisbane, Australia for giving their time

to this study. We would also like to acknowledge the medical staff who collected the injury data. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported that there is no funding associated with the work featured in this article.

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Data availability statement

The datasets generated during and/or analysed during the current study are not publicly available due to case numbers and project ethical requirements but may be available from the corresponding author who may be contacted for further information.

Ethics approval and informed consent

Ethics approval and informed consent were obtained for this study. This study was approved by the University ethics committee (HREC 2020/113).

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